# CASH FOR CLIMATE ADAPTATION IN UGANDA: EXPERIMENTAL DESIGN AND RANDOMIZATION*

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Abstract. [Abstract placeholder.]

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## 1. Introduction

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## 2. Data

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## 3. Empirical Strategy

This section describes the experimental design, the construction of randomization clusters, and the power calculations that inform the study's ability to detect treatment effects.

3.1. **Experimental Design.** The program is implemented across 134 settlements in Bulambuli and Amuria Districts, Uganda, reaching approximately 11,801 recipients. All eligible adults in selected settlements receive a $644 USD lump-sum cash transfer ($50 token followed by $594 lump sum). Settlements in Amuria are substantially larger than those in Bulambuli: using parish-level population data from the Uganda Bureau of Statistics (UBOS-NPHC 2024), we estimate that Amuria settlements average approximately 222 households, while Bulambuli settlements average approximately 29 households—a ratio of nearly 8:1. Despite having only 16 settlements, Amuria accounts for roughly half of the program's total households (3,558 of 6,928).

The evaluation uses a three-arm design with randomization at the cluster level (see Section 3.3 for cluster construction):

- **Group 1**: Standard Large Lump Sum (LLS) program with a default "Community Empowerment" Baraza. Cash delivered early in the rollout (June–November 2026). Serves as control for the plus comparison and as treatment for the cash comparison.

- **Group 2**: Same cash transfer and timeline as Group 1, but with a modified Baraza framing transfers as supporting climate adaptation, plus household-level randomized components (historical climate information and adaptation aspirations planning).
- **Group 3**: Standard LLS with default Baraza, delivered later in the rollout. Serves as control for the cash comparison.

The design addresses two causal questions. First, what is the effect of unconditional cash transfers on climate-adaptive behaviors (Group 1 vs. Group 3)? Second, what is the additional effect of climate-focused "plus" components (Group 2 vs. Group 1)?

3.2. **Hypotheses and Estimation Equations.** The design addresses three questions, each with a distinct estimation equation.

Question 1: Cash effect (Group 1 vs. Group 3).

$$Y_{iv} = \alpha + \beta D_v^{(1)} + \gamma Y_{iv,\text{baseline}} + \delta X_{iv} + \varepsilon_{iv} \tag{3.1}$$

where $Y_{iv}$ is the outcome for household $i$ in village $v$ at follow-up (e.g., 3-month livelihood investment), $D_v^{(1)} = 1$ if the village is in Group 1 (early cash) and 0 if in Group 3 (late cash), $Y_{iv,\text{baseline}}$ is the same outcome measured at baseline, and $X_{iv}$ are household characteristics. We cluster standard errors at the village level and do not include village fixed effects because treatment is assigned at the village level. Power comes from the number of clusters (approximately 9 per arm).

Question 2: Baraza framing effect (Group 1 vs. Group 2c).

$$Y_{iv} = \alpha + \beta D_v^{(2c)} + \gamma Y_{iv,\text{baseline}} + \delta X_{iv} + \varepsilon_{iv} \tag{3.2}$$

We cluster standard errors at the village level and do not include village fixed effects because Baraza type varies at the village level and fixed effects would absorb that variation. This comparison has similar power to Q1 (approximately 9 vs. 3 clusters).

Question 3: Plus components (Group 2a vs. Group 2b vs. Group 2c).

$$Y_{iv} = \alpha + \beta D_i^{(2a)} + \gamma Y_{iv,\text{baseline}} + \delta X_{iv} + \mu_v + \varepsilon_{iv} \tag{3.3}$$

We include village fixed effects $\mu_v$ because randomization is at the household level within villages. Standard errors are clustered at the village level. Power comes from the number of households (∼300 per comparison), allowing detection of smaller effects (∼0.2–0.3 SD).

3.3. **Cluster Construction and Randomization.** The two study districts are geographically separated by approximately 98 kilometers, eliminating any concern about cross-district spillovers. Figure 1 displays all 134 settlements colored by district.

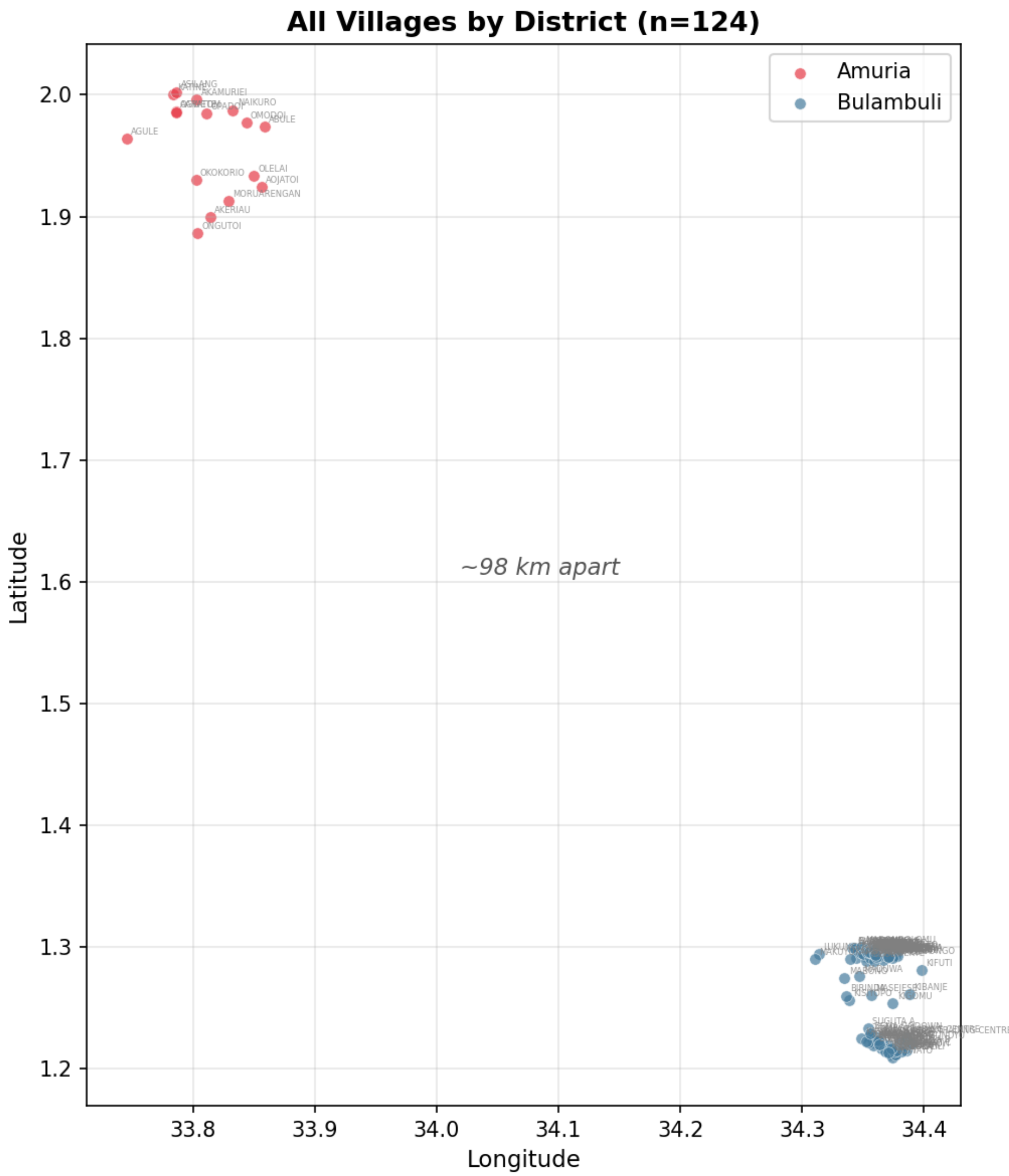


Figure 1. All study settlements by district. The two districts are approximately 98 km apart.

Figure 2 displays settlement locations with dot size proportional to estimated households, illustrating the stark size asymmetry between districts.

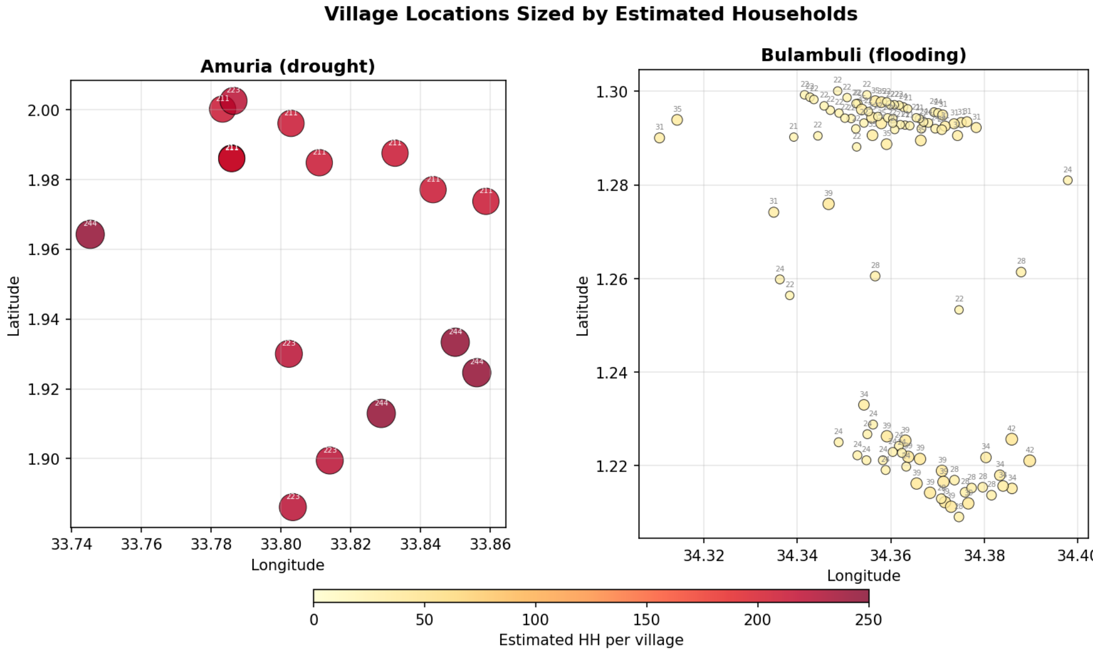


Figure 2. Settlement locations sized by estimated households per settlement (UBOS 2024 parish-level data).

3.3.1. *The Spillover Problem.* A naive village-level randomization assigns each of the 134 settlements independently to one of the three arms. While this maximizes the number of randomization units (and thus statistical power), it creates a severe spillover problem in Bulambuli, where settlements are densely packed within 1–2 kilometers of each other. Figure 3 illustrates a simulated village-level randomization: neighboring settlements in Bulambuli are assigned to different arms, meaning that recipients in early-cash settlements and delayed-cash settlements share local markets, trade networks, and social ties. Treatment effects from cash settlements would contaminate control settlements, biasing the estimated cash effect toward zero.

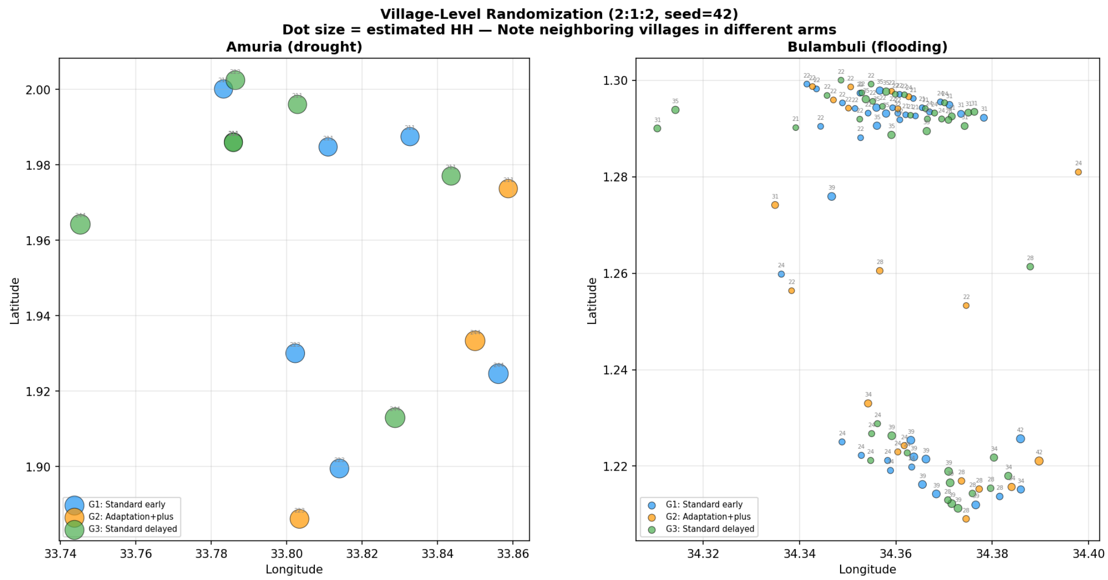


FIGURE 3. Simulated village-level randomization (2:1:2 ratio). Note the extensive mixing of arms among neighboring villages in Bulambuli.

3.3.2. *Geographic Clustering.* To address spillovers, we construct randomization clusters using hierarchical agglomerative clustering (complete linkage) on settlement GPS coordinates. Under complete linkage, two settlements are merged into the same cluster only if the maximum pairwise distance between any two settlements in the resulting cluster is below the distance threshold. We evaluate thresholds of 1, 2, and 3 kilometers.

Figure 4 shows how the number of clusters varies with the distance threshold. Bulambuli settlements collapse rapidly from 46 clusters at 0.5 km to 8 clusters at 3 km, reflecting their geographic density. Amuria settlements remain relatively stable at 9–14 clusters, reflecting their natural spatial separation.

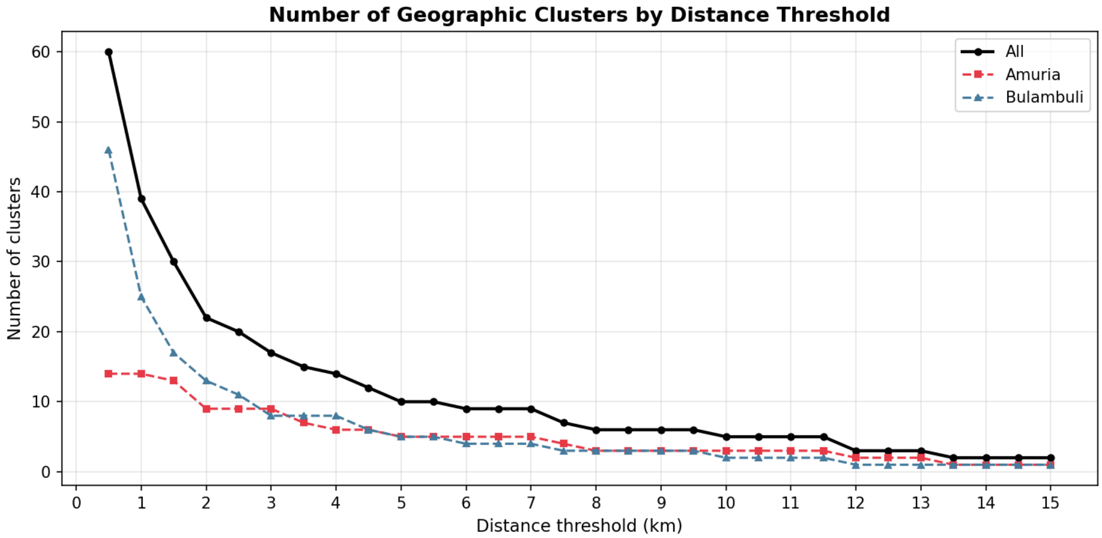


FIGURE 4. Number of geographic clusters as a function of the distance threshold.

Figures 5–7 display the resulting randomization under geographic clustering at 1, 2, and 3 kilometer thresholds. As the threshold increases, the spatial separation between arms improves, but the number of randomization units decreases.

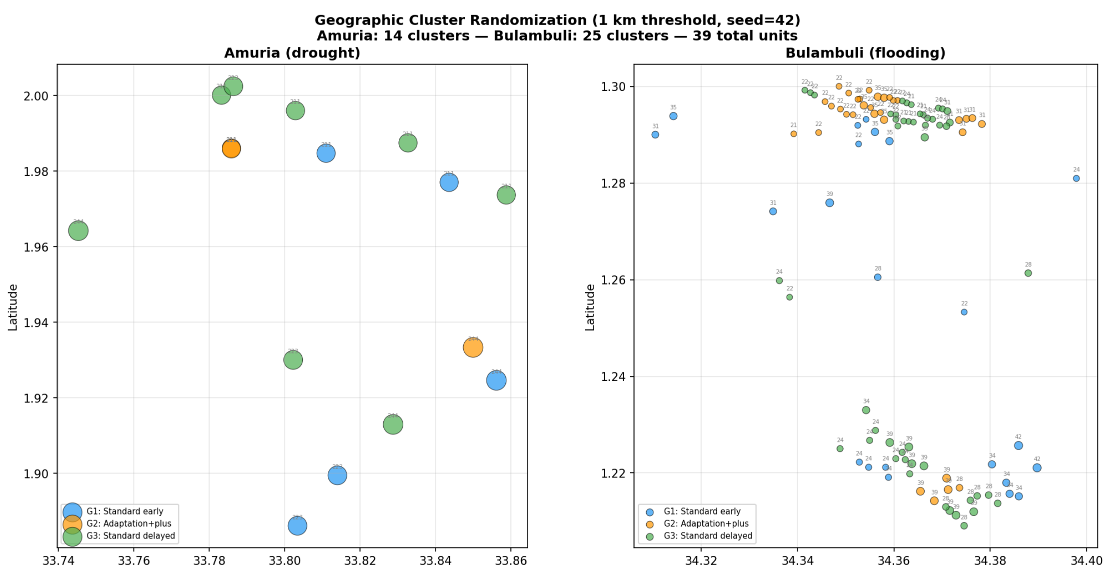


FIGURE 5. Geographic cluster randomization at 1 km threshold.

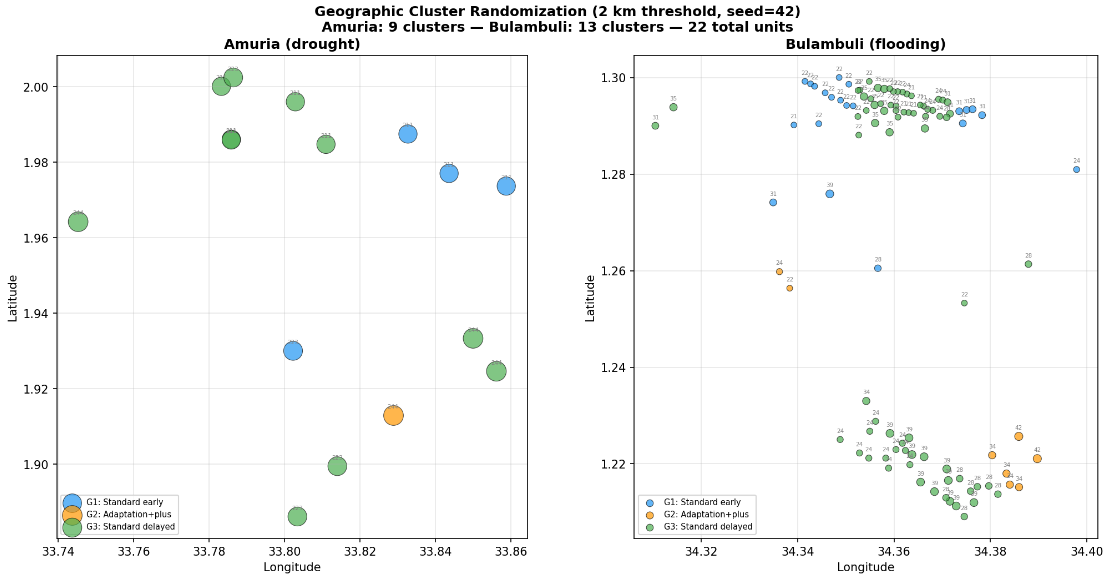


FIGURE 6. Geographic cluster randomization at 2 km threshold.

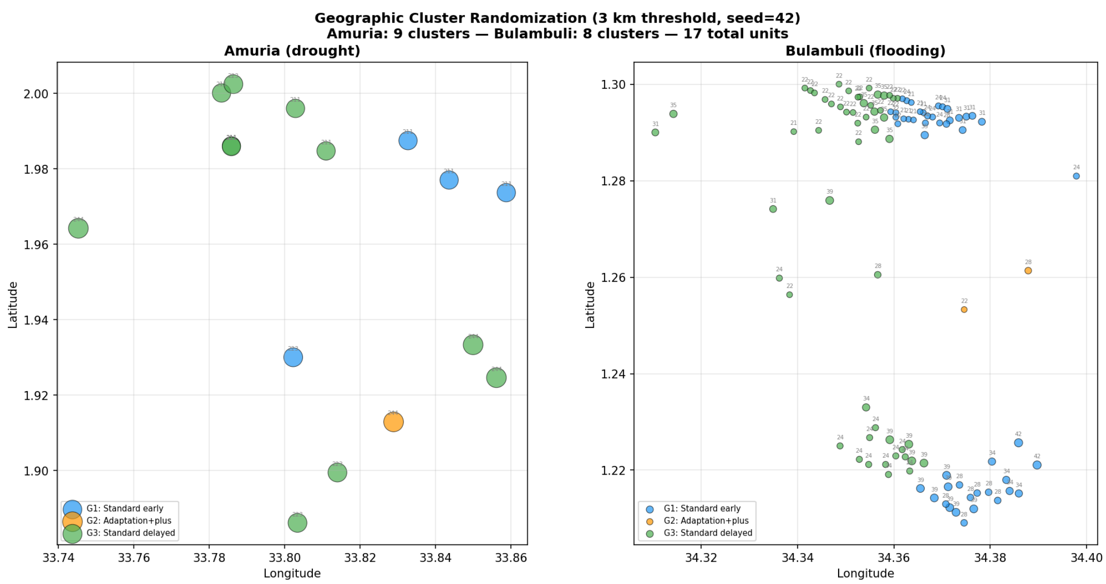


FIGURE 7. Geographic cluster randomization at 3 km threshold.

3.3.3. *Manual Cluster Adjustment.* We adopt the 3 km threshold as the base and apply manual adjustments informed by visual inspection of the village map. In Amuria, where villages are naturally spread out, we split three multi-village clusters (C9, C10, C12 in the base clustering) into individual or smaller units, since the constituent villages are 1.5–1.9 km apart with no shared market catchment. In Bulambuli, we split one mid-range cluster (C6) into a southern pair and a northern group separated by approximately 2 km. The resulting assignment yields 22 randomization clusters: 13 in Amuria and 9 in Bulambuli. Figure 8 displays the final cluster assignment.

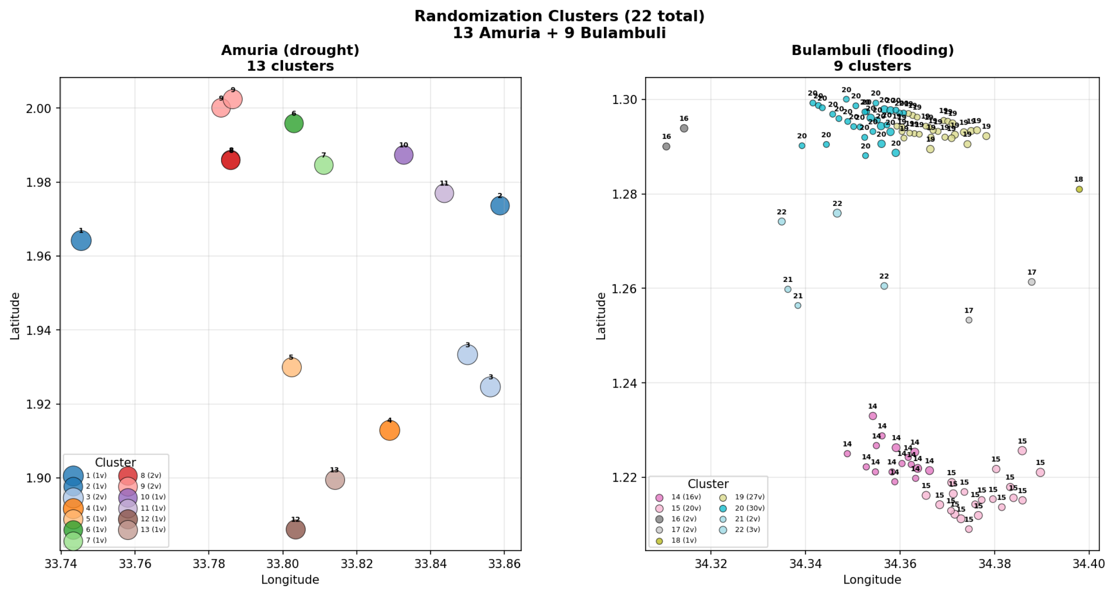


FIGURE 8. Final cluster assignment: 22 clusters (13 Amuria, 9 Bulambuli).

3.3.4. *Randomization.* We randomize the 22 clusters into three arms using a 2:1:2 allocation ratio, stratified by district. Within each district, clusters are randomly assigned to Group 1 (standard LLS, early cash), Group 2 (adaptation Baraza plus household-level pluses), or Group 3 (standard LLS, delayed cash). The randomization is performed once with a fixed seed and documented for reproducibility. Figure 9 displays the resulting assignment.

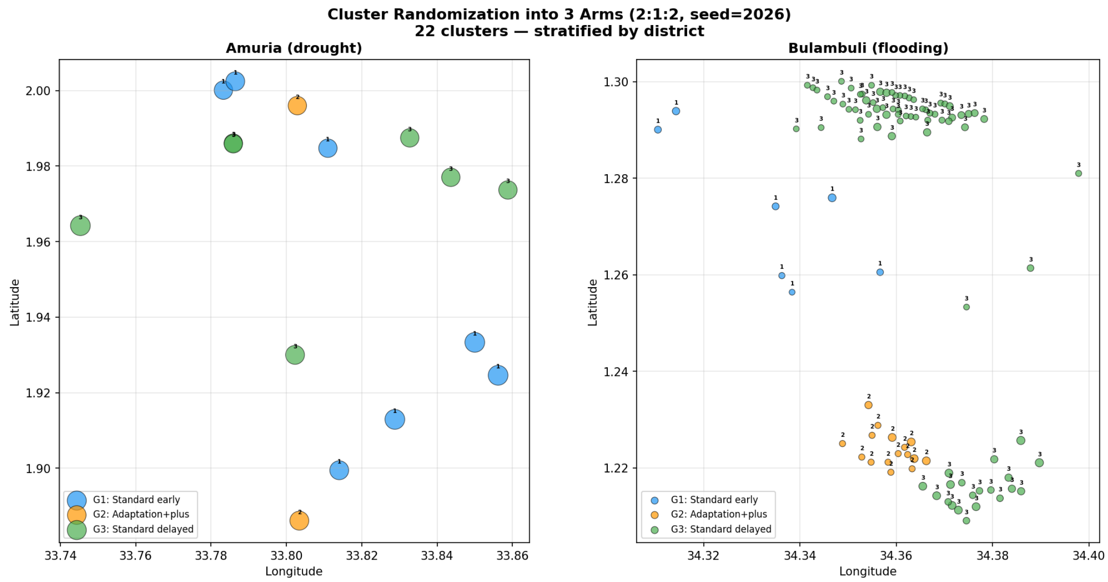


Figure 9. Final randomization of 22 clusters into three arms (2:1:2 ratio, stratified by district).

3.4. **Power Calculations.** We evaluate the minimum detectable effect (MDE) for the primary comparison (Q1: cash vs. no cash, Group 1 vs. Group 3) under the cluster-randomized design. The MDE depends on the number of clusters per arm, the number of households surveyed per cluster, the intra-cluster correlation coefficient (ICC), and the baseline-endline correlation from the panel design.

3.4.1. *Parameters.* We draw on two literature reviews to calibrate the power calculations (see Appendix C for full details). For the ICC, a synthesis of empirical estimates from cash transfer evaluations in East Africa—including Haushofer and Shapiro (2016), McIntosh and Zeitlin (2022), Egger et al. (2022), and Aggarwal et al. (2023)—yields a recommended range of 0.05 (low), 0.10 (medium), and 0.15 (high) for livelihood investment outcomes. The most directly relevant Uganda estimate is ICC = 0.07 from ICAN-ICAR (2025). For baseline outcome parameters, we use a central estimate of 75,000 UGX per month for livelihood investment spending (SD = 112,500 UGX), calibrated to extreme-poor populations in Eastern Uganda using the SAGE evaluation

(Government of Uganda, 2012) and Blattman et al. (2014). For the binary diversification outcome, we assume a baseline rate of 20%, informed by non-agricultural participation rates in Aggarwal et al. (2023) and McIntosh and Zeitlin (2022).

The panel design (baseline and endline surveys of the same households) provides an ANCOVA gain through the baseline-endline correlation, which we set at $\rho = 0.5$. We survey 30% of households per settlement, yielding a weighted average of approximately 16 households surveyed per cluster (reflecting the mix of large Amuria settlements and small Bulambuli settlements).

3.4.2. *Minimum Detectable Effects.* Figure 10 displays the MDE as a function of the number of clusters per arm, for both the continuous outcome (livelihood investment, left panel) and the binary outcome (income diversification, right panel). Horizontal reference lines indicate effect sizes observed in prior GiveDirectly evaluations. The gray shaded region marks the village-level design (56 clusters per arm), and the blue shaded region marks our cluster-based design (approximately 9 clusters per arm).

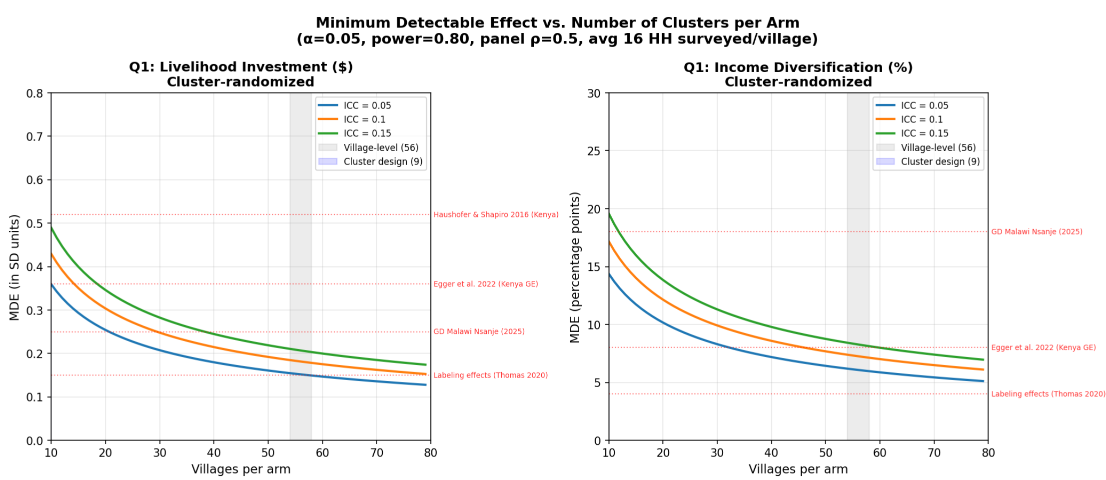


FIGURE 10. Minimum detectable effect vs. number of clusters per arm. Gray region: village-level randomization (56 per arm). Blue region: cluster-based design (9 per arm, 22 clusters total). Horizontal lines: effect sizes from prior cash transfer evaluations.

At 56 clusters per arm (village-level randomization), the MDE ranges from 0.15 to 0.21 SD for the continuous outcome, well below the 0.25–0.52 SD effects observed in prior GiveDirectly programs (Haushofer and Shapiro, 2016; Egger et al., 2022). At 9 clusters per arm (our cluster-based design), the MDE increases to approximately 0.30–0.45 SD, which remains within the range of plausible cash transfer effects but leaves less margin for smaller-than-expected impacts. The cluster-based design sacrifices statistical power in exchange for clean identification by eliminating within-cluster spillovers.

For the binary diversification outcome, the village-level design can detect changes of 6–8 percentage points, while the cluster design can detect changes of approximately 10–17 percentage points. The GiveDirectly Malawi program observed an 18 percentage point increase in business ownership (Aggarwal et al., 2023), suggesting that effects of this magnitude would be detectable under either design.

For Q2 (plus effect), randomization is at the household level within Group 2 settlements. With approximately 149 households per sub-arm, the MDE is 0.28 SD for the continuous outcome and 11.2 percentage points for the binary outcome. Given that labeling and messaging effects in the literature are typically 0.10–0.25 SD (Thomas et al., 2020), Q2 is likely underpowered for small effects and should be interpreted as exploratory.

## Appendix A. Appendix: Program Design Details

A.1. **Program Overview.** The Quadrature Climate Foundation (QCF) funds a program with GiveDirectly to test whether unconditional cash transfers can support climate adaptation in Uganda. The program targets 11,801 recipients across 134 settlements in Bulambuli District (flooding, waterlogging) and Amuria District (drought). Each recipient receives $644 USD in two tranches: a $50 token transfer after enrollment, followed by a $594 lump sum several weeks later.

A.2. **Theory of Change.** Cash relaxes liquidity and capital constraints, enabling households to make forward-looking livelihood investments that support ex-ante climate adaptation. Households may allocate transfers toward: (1) expanding existing livelihoods, (2) diversifying into new income-generating activities, (3) making existing livelihoods more resilient to climate risk, (4) general consumer durables, or (5) climate-protective durable assets. Channels (1), (2), and (4) correspond to the standard LLS economic pathway; channels (3) and (5) correspond to the climate adaptation pathway.

A.3. **Indicators.** The evaluation collects 15 standard LLS indicators (food expenditure, household consumables, education, consumer durables, savings/debt, livelihood investments, healthcare, food insecurity, dietary diversity, psychological well-being, income-generating activity, hours worked, asset ownership, and debt/savings) and 10 non-standard climate adaptation indicators organized into four domains:

(1) **Expectations and Constraints:** Perceived ability to prepare for climate risks; priority support needed for adaptation.
(2) **Allocation Choices:** Livelihood investment decomposed into expansion, diversification, and resilience categories.
(3) **Structural Adjustment:** Climate-related occupational change; climate-related migration.

(4) **Spillovers and System Effects:** Individual investments with community spillovers; positive coping in response to shock; maladaptive coping behavior.

A.4. **M&E Design.** Surveys are administered to 30% of recipients (∼3,540 households) at baseline and endline, using a panel design. Baseline data collection occurs at enrollment before the first transfer. Endline occurs approximately three months after the final transfer delivery. Surveys are conducted in person and take approximately 30 minutes.

A.5. **Limitations.**

(1) No direct shock tracking between baseline and endline.
(2) No non-recipient comparison group for the within-group analysis; the staggered design (Group 1 vs. Group 3) provides the cash comparison.
(3) Limited cross-program comparability.
(4) Baraza labeling is integrated into the program design and is not separately estimated.

## Appendix B. Appendix: MELP Key Information

The research questions and three-arm design are described in Sections 3.1 and 3.2.

B.1. **M&E Survey Design.**

- 30% of recipients surveyed at baseline and endline (∼3,540 households).
- Panel design: same households at both waves.
- Baseline at enrollment (before first transfer); endline ∼3 months after final transfer.
- In-person surveys, ∼30 minutes.

B.2. **Standard LLS Indicators (1–15).**

(1) Spend on food
(2) Spend on household consumables

(3) Spend on children's education

(4) Spend on consumer durables

(5) Save or repay debt

(6) Spend on livelihood investments

(7) Spend on healthcare

(8) % experiencing moderate/severe food insecurity

(9) % consuming $\geq$5 food groups

(10) Psychological well-being score

(11) % engaged in income-generating activities

(12) Average hours worked per week

(13) % reporting increased income-generating activity

(14) % owning productive/non-productive assets

(15) Total value of debt and savings

B.3. **Non-Standard Climate Adaptation Indicators. 0. Experienced Climate Shock** — % reporting a climate-related shock and timing.

**I. Expectations & Constraints**

(1) Perceived ability to prepare for climate risks

(2) Priority support needed for climate adaptation

**II. Allocation Choices**

(3) Livelihood investment — expansion (continuing pre-existing activities)

(4) Livelihood investment — diversification (new income-generating activities)

(5) Livelihood investment — resilience (protective improvements to existing livelihoods)

**III. Structural Adjustment**

(6) Climate-related occupational change

(7) Climate-related migration

**IV. Spillovers & System Effects**

(8) Individual investments with community spillovers

(9) Positive coping in response to shock

(10) Maladaptive coping behavior (charcoal burning, bush burning, wetland encroachment)

B.4. **Reporting Timeline.**

- First report: October 2026
- Final report: October 2027

## Appendix C. Appendix: Power Calculation Parameters

This appendix documents the empirical sources for the ICC values and baseline outcome parameters used in the power calculations (Section 3.4). M&E surveys are administered to 30% of recipients in each settlement, yielding the household counts per cluster used throughout.

C.1. **Intra-Cluster Correlation Coefficients.** The ICC determines the proportion of total outcome variance attributable to between-cluster differences. Table 1 summarizes empirical estimates from cash transfer evaluations and household surveys in rural Sub-Saharan Africa.

TABLE 1. Empirical ICC estimates from rural Sub-Saharan Africa

| Domain | Outcome | ICC | Type | Source |
|---|---|---|---|---|
| Consumption | Total consumption | 0.25 | Reported | World Bank (2019) |
| Consumption | Food expenditure | 0.16 | Reported | World Bank (2019) |
| Consumption | Monthly expenditure | 0.06 | Derived | Haushofer & Shapiro (2016) |
| Consumption | Monthly expenditure | 0.05 | Derived | McIntosh & Zeitlin (2022) |
| Consumption | Annual expenditure | 0.07 | Derived | ICAN-ICAR (2025), Uganda |
| Food security | Dietary diversity | 0.03 | Derived | McIntosh & Zeitlin (2022) |
| Food security | Not worried about food | 0.02 | Derived | Psychology of Poverty (2021) |
| Food security | Food security index | 0.04 | Derived | Haushofer & Shapiro (2016) |
| Livelihood | Livestock (TLU) | 0.12 | Derived | Psychology of Poverty (2021) |
| Livelihood | Irrigation adoption | 0.18 | Reported | Asian J. Agric. Ext. (2025) |
| Livelihood | Business entry | 0.07 | Derived | McIntosh & Zeitlin (2022) |
| Assets | Non-land asset index | 0.11 | Derived | Haushofer & Shapiro (2016) |
| Assets | Productive asset value | 0.09 | Derived | McIntosh & Zeitlin (2022) |
| Income | Non-agricultural income | 0.06 | Derived | Aggarwal et al. (2023) |
| Income | Enterprise revenue | 0.10 | Reported | Egger et al. (2022) |
| Income | Wage labor participation | 0.05 | Derived | McIntosh & Zeitlin (2022) |

Based on this synthesis, we adopt the following ranges: 0.05–0.15 for livelihood investment and income diversification (primary outcomes), and 0.01–0.05 for food security (secondary). The most directly relevant Uganda estimate is ICC = 0.07 from ICAN-ICAR (2025).

Egger et al. (2022) document substantial spatial interference in the GiveDirectly Kenya trial, where treatment in one village affected enterprise outcomes in neighboring untreated villages. This motivates our geographic clustering approach (Section 3.3).

C.2. **Baseline Outcome Means and Standard Deviations.** Table 2 summarizes baseline estimates from national surveys and experimental evaluations, prioritizing extreme-poor populations in Eastern Uganda and comparable settings.

Table 2. Empirical baseline means and standard deviations

| Domain | Variable | Mean | SD | Setting | Source |
|---|---|---|---|---|---|
| Consumption | HH monthly exp. (UGX) | 372,128 | 154,611 | Uganda (rural) | UNHS 2019/20 |
| Consumption | HH monthly cons. (UGX) | 40,599 | 16,867 | Uganda (extreme poor) | SAGE 2012 |
| Consumption | HH monthly cons. (USD PPP) | 158 | 296 | Kenya (rural) | Haushofer (201 |
| Livelihood | Monthly earnings (USD) | 20 | 34 | Uganda (north) | Blattman (2014 |
| Livelihood | Non-ag income (USD PPP) | 15.28 | 18.10 | Malawi (rural) | Aggarwal (2023 |
| Livelihood | HH revenue (USD PPP) | 49 | 85 | Kenya (rural) | Haushofer (201 |
| Food security | Severe food insecurity (%) | 60.2 | 0.49 | Uganda (eastern) | Cambridge (202 |
| Food security | Not worried about food (%) | 15.0 | 0.36 | Malawi (rural) | Psych. Poverty |
| Assets | Non-land assets (USD PPP) | 495 | 580 | Kenya (rural) | Haushofer (201 |
| Assets | Livestock value (USD PPP) | 127 | 230 | Kenya (rural) | Haushofer (201 |

Our target population (85% multidimensionally poor in Bulambuli) is closer to the SAGE extreme-poor sample than to the Kenya GiveDirectly or national Uganda averages. We adopt a central estimate of 75,000 UGX/month for livelihood investment spending (SD = 112,500 UGX, CV = 1.5), a baseline diversification rate of 20%, and a food insecurity rate of 60%. These are working assumptions that will be updated with baseline data.

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